

Dealing with Uncertainty

Probability Theory

- Uncertainty can be represented as probabilities.
- If event A occurs N_A times out of a total of N occasions, then the probability of event A is

$$P(A) = N_A / N$$

Prior or Unconditional Probabilities

- When we have **no other information**, it is appropriate to use the prior or unconditional probability $p(A)$ of event A happening.
 - $P(\text{sunny weather}) = 0.72$
 - $P(\text{Lift stops}) = 0.1$
- Let A be a proposition, $P(A)$ denotes the unconditional probability that A is true.
- Example: if M denotes the proposition that a particular person is male, then $P(M)=0.5$ means that without any other information, the probability of that person being male is 0.5 (a 50% chance).
- Alternatively, if a population is sampled, then 50% of the people will be male.
- However, with additional information (e.g. that the person is a Mechanical Engg. student), the “posterior probability” will likely be different. (as mostly boys prefer mechanical Engg. It can be 80%.

Conditional or Posterior Probabilities

If we have some information about the problem domain we need to use conditional or posterior probabilities

$$p(A | B) = \text{"probability of A given B"}$$

Combining this with a prior probability by the *product rule* gives the *joint probability*

$$p(A \wedge B) = p(A | B) \cdot p(B)$$

Reversing this gives the conditional probabilities in terms of unconditional probabilities

$$p(A | B) = \frac{p(A \wedge B)}{p(B)}$$

Probability distribution gives probabilities of all possible values of a random variable.

Weather is one of <sunny,rain,cloudy,snowy>

$\mathbf{P}(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$

(normalized, i.e., sum= 1)

Each random variable X will have a domain of possible values

$D_X = \{x_1, x_2, \dots, x_n\}$ that it can take.

$$0 \leq p(x_i) \leq 1 \quad , \quad \sum_{i=1}^n p(x_i) = 1 \quad , \quad p(\neg x_i) = 1 - p(x_i)$$

Probabilities over many independent domains:

if we have independent variables X and Y, we have

$p(x_1, y_2) \equiv p(x_1 \wedge y_2) = p(x_1) \cdot p(y_2)$ //product of 2 probabilities

The probability of a disjunction is given by

$$p(A \vee B) = p(A) + p(B) - p(A \wedge B) .$$

If A and B are independent, then

$$p(A|B) = p(A) , \text{ and } p(A \wedge B) = p(A).p(B)$$

Possible approaches to handle uncertainty

- Conditional planning
 - Plan to obtain information
 - **Demerit:** This is expensive because one has to plan for many unlikely cases
- Monitoring/Replanning
 - Assume normal states & outcomes
 - Check progress during execution, replan if necessary
 - **Demerit:** unanticipated outcomes may lead to failure

Example: Dealing with Uncertainty using Probability

- Let action A_t represent leaving for the airport t minutes before the departure of the flight
- Will A_t get me to the airport on time?

Problems:

1. Partial observability (road condition, unpredictable traffic, etc.)
2. Noisy sensors (incorrect traffic report)
3. Uncertainty in action outcomes (turn key, car doesn't start, etc.)

Methods for handling uncertainty

Probability:

Given the available evidence,

A_{90} will get me to the airport on time with probability 0.95

Subjective or Bayesian probability:

$P(A_{90} \text{ succeeds} \mid \text{no reported accidents}) = 0.97$

Probabilities of propositions change with new evidence:

$P(A_{90} \mid \text{no reported accidents, 5 a.m.}) = 0.99$

Decision under uncertainty

- Suppose I believe the following:
 $P(A_{30} \text{ gets me there on time} \mid \dots) = 0.05$
 $P(A_{60} \text{ gets me there on time} \mid \dots) = 0.70$
 $P(A_{100} \text{ gets me there on time} \mid \dots) = 0.95$
 $P(A_{1440} \text{ gets me there on time} \mid \dots) = 0.9999$
- Which action to choose?
Depends on my preferences for missing flight
- ***Utility theory*** is used to represent and infer preferences
- ***Decision theory = utility theory + probability theory***

Joint probability distribution

Joint probability distribution for a set of variables gives values for each possible assignment to all the variables

$P(\text{Toothache}, \text{Cavity})$ is a 2 by 2 matrix.

	Toothache=true	Toothache = false
Cavity=true	0.04	0.06
Cavity=false	0.01	0.89

Conditional Probabilities

- Conditional or posterior probabilities

e.g., $P(\text{Cavity} \mid \text{Toothache}) = 0.8$

What is the probability of having a cavity given that the patient has a toothache?

(Ex: $P(\text{Male} \mid \text{Mechanical Engg. student}) = ??$)

- If we know more, e.g., dental probe catches, then we have
 $P(\text{Cavity} \mid \text{Toothache}, \text{Catch}) = .9$ (say)
- If we know even more, e.g., Cavity is also given, then we have
 $P(\text{Cavity} \mid \text{Toothache}, \text{Cavity}) = 1$

Conditional Probabilities

- **Conditional or posterior probabilities**

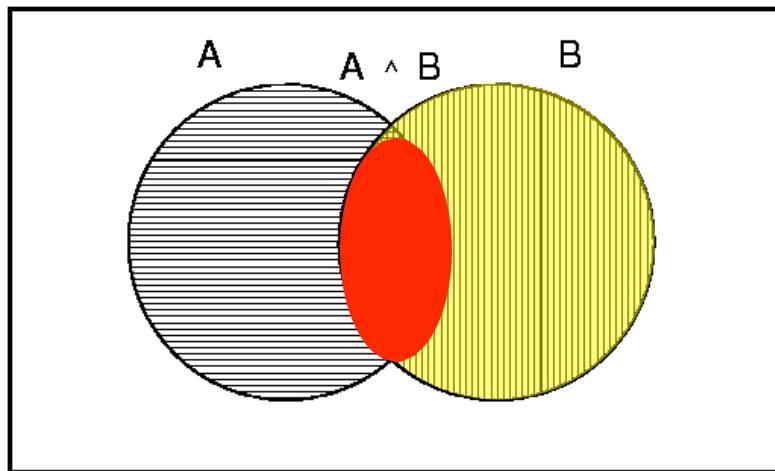
e.g., $P(\text{Cavity} \mid \text{Toothache}) = 0.8$

What is the probability of having a cavity given that the patient has toothache?

- Definition of **conditional probability**:

$$P(A|B) = P(A, B) / P(B) \quad \text{if } P(B) \neq 0$$

True



I.e., $P(A|B)$ means, our universe has shrunk to the one in which B is true.

$P(A|B)$ is the ratio of the red area to the yellow area.

Conditional Probabilities

- **Conditional** or posterior probabilities

e.g., $P(\text{Cavity} \mid \text{Toothache}) = 0.8$

What is the probability of having a cavity given that the patient has a toothache?

- Definition of **conditional probability**:

$$P(A|B) = P(A, B) / P(B) \quad \text{if } P(B) \neq 0$$

	Toothache=true	Toothache = false
Cavity=true	0.04	0.06
Cavity=false	0.01	0.89

$$P(\text{Cavity}|\text{Toothache}) = .04/ (.04+.01) = 0.8$$

Conditional probability cont.

Definition of conditional probability:

$$P(A | B) = P(A, B) / P(B) \quad \text{if } P(B) \neq 0$$

Product rule gives an alternative formulation:

$$P(A, B) = P(A | B)P(B) = P(B|A)P(A)$$

Example:

$$P(\text{Male}) = 0.5$$

$$P(\text{Mech. Engg. Student}) = 0.0037$$

$$P(\text{Male} | \text{Mech. Engg. Student}) = 0.9$$

What is the probability of being a male Mech. Engg. Student?

$$P(\text{Male, Mech. Engg. Student})$$

$$= P(\text{Male} | \text{Mech. Engg. Student})P(\text{Mech. Engg. Student})$$

$$= 0.0037 * 0.9 = 0.0033$$

Bayes Rule

From product rule $P(A \wedge B) = P(A|B)P(B) = P(B|A)P(A)$, we can obtain Bayes' rule (also known as Bayes' Law or Bayes' Theorem)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

This simple equation underlies all modern AI systems for probabilistic inference

For assessing diagnostic probability from causal probability:

e.g. Cavities as the cause of toothaches

Sleeping late as the cause for missing class

Meningitis as the cause of a stiff neck

$$P(Cause | Effect) = \frac{P(Effect | Cause)P(Cause)}{P(Effect)}$$

Bayes Rule: Example

$$P(Cause | Effect) = \frac{P(Effect|Cause)P(Cause)}{P(Effect)}$$

Let M be meningitis, S be stiff neck

$$P(M)=0.0001$$

$$P(S)= 0.1$$

$$P(S | M) = 0.8$$

$$P(M | S) = \frac{P(S | M)P(M)}{P(S)} = \frac{0.8 * 0.0001}{0.1} = 0.0008$$

A generalized Bayes Rule

- More general version conditionalized on some background evidence E

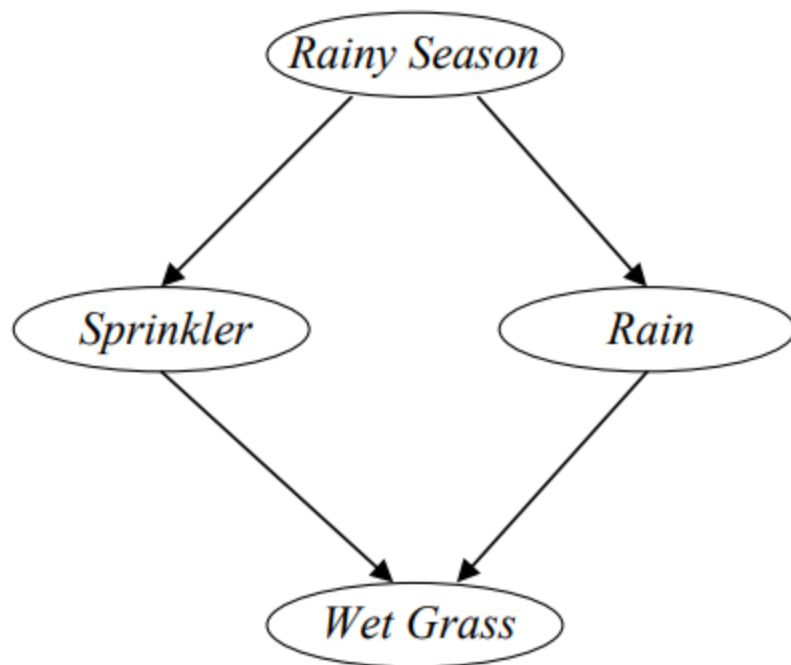
$$P(A | B, E) = \frac{P(B | A, E)P(A | E)}{P(B | E)}$$

Bayesian Networks

- A Bayesian Network is a directed acyclic graph that represents **causality relationships**.
- The directed links between the nodes in the graph indicate direct influence. Missing links (usually much greater in number) correspond to the conditional independencies.
- Each node has either a prior probability, or a conditional probability table that quantifies the effects of its parents (i.e. the nodes that influence it).
- We need a mechanism for computing the influence of any node on any other. We also need to ensure that the probabilities are transmitted correctly.

Example of Bayesian Network

- Consider the simple example of grass that may be wet because of rain or sprinkler:



Attribute	Probability
$p(Wet Sprinkler,Rain)$	0.95
$p(Wet Sprinkler,\neg Rain)$	0.9
$p(Wet \neg Sprinkler,Rain)$	0.8
$p(Wet \neg Sprinkler,\neg Rain)$	0.1
$p(Sprinkler RainySeason)$	0.0
$p(Sprinkler \neg RainySeason)$	1.0
$p(Rain RainySeason)$	0.9
$p(Rain \neg RainySeason)$	0.1
$p(RainySeason)$	0.5